Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of AI-Based Process Recognition and Automation Technologies for Enhancing Quality Uniformity and Verifying Structural Performance of Pultruded Composite Prototypes**			
Korea's 12 major national strategic technology fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Machinery and manufacturing			
	Medium category	Small category	Medium category	Small category
	Manufacturing process automation	***Process monitoring and quality control***		
Proposed technology concept	**Core Concept** Pultrusion is a representative continuous composite manufacturing process that enables economical production of structural composite members. It has strong potential for application across various industrial sectors—including construction, shipbuilding, marine structures, and electrical/industrial components. However, despite the numerous process variables involved, such as roving alignment, fiber tension, resin viscosity and impregnation conditions, and die temperature, there is a structural limitation: the lack of real-time, quantitative methods to assess product quality during processing. This limitation results in quality fluctuations. In particular, nonuniform roving alignment and insufficient resin impregnation directly lead to void formation, fiber volume fraction deviations, and reductions in structural performance—factors that represent major barriers to deploying pultruded composites as high-reliability structural materials.			

Conventional pultrusion processes rely on post-process material testing or cross-sectional analysis to detect such defects, making it difficult to intervene in real time and challenging to ensure production repeatability.

The core concept of this technology is to recognize roving alignment and resin impregnation conditions in real time during the pultrusion process and to automatically stabilize process conditions based on this information, thereby quantitatively improving product quality consistency and structural performance. To achieve this, AI-based process recognition is introduced—not for the purpose of enhancing AI capabilities themselves, but as a supportive tool for objective evaluation of process states. This differentiates the proposed approach from conventional AI-centered manufacturing projects.

In essence, the technical foundation lies in pultrusion process technology, while AI-based recognition is used as a tool to quantitatively interpret complex process-state interactions and reflect them in process control. Through this approach, the project aims to systematically analyze how variations in process conditions affect product quality and to establish a pultrusion process technology capable of maintaining stable quality and structural performance even during repeated production.

Key Functions and Potential Applications

- Key Functions

  The technology to be developed in this project follows a stepwise framework consisting of process recognition-evaluation-control-verification across the pultrusion process. The major functions are as follows.

  - Pultrusion Process State Recognition

    Sensor- and vision-based data acquisition techniques are applied to recognize roving alignment and resin impregnation conditions in real time. By detecting alignment quality, uniformity, local twisting, and deviations during the process, the system overcomes traditional reliance on visual inspection and post-analysis, enabling real-time monitoring of process states.

    For resin impregnation, the system identifies changes in impregnation uniformity and potential dry-spot formation under varying process conditions. This allows early detection of the most frequent causes of quality degradation in pultrusion processes.

  - AI-based Process Assessment and Automated Control

    This function develops AI-based process recognition and automation technologies that assess quality conditions in real time based on collected roving alignment and resin impregnation data, and link the assessment results to process control.

    The AI interprets complex process-state variations quantitatively and distinguishes between normal and abnormal states. The objective is not simple image recognition or data analysis but the establishment of quality-oriented decision criteria.

    Because pultrusion quality is affected by many interdependent variables—roving tension, resin viscosity, process speed, die temperature—simple monitoring of individual variables is insufficient. Therefore, a decision logic will be developed that

integrates multi-source data to detect early-stage abnormalities, such as roving misalignment or impregnation deviations, and to predict their impact on structural performance.

Based on the assessed process state, automated control adjusts key process parameters such as roving tension, resin feed conditions, and pulling speed. This quantitative stabilization system reduces operator dependency and minimizes quality fluctuations. The goal is not full autonomy but a field-applicable automation approach that ensures repeatability and quality consistency, enabling stable production of structural pultruded components.

- Prototype Fabrication and Structural Performance Verification

  Pultruded composite prototypes are manufactured under optimized process conditions, and structural performance tests (tensile, flexural, etc.) are conducted to verify process effectiveness. Quantitative indicators such as mechanical property variations and fiber volume fraction uniformity are evaluated to objectively demonstrate improvements before and after applying AI-based process recognition and automation.

  This confirms that the project generates not only automation technologies but also tangible improvements in product quality and structural performance.

- Potential Applications
  - Industrial Sectors

    The AI-assisted pultrusion process recognition and automation technology developed in this project is not limited to specific products. It is a process-level technology applicable to various composite structural components requiring both mechanical performance and quality consistency.

    In construction and infrastructure applications such as bridge components, railings, decks, and reinforcement structures, the lightweight and corrosion-resistant characteristics of pultruded composites provide strong applicability. With quantitatively validated quality consistency and structural performance, these composites can replace conventional metallic structural materials.

    In shipbuilding, marine, and offshore structures, demand for composites continues to increase due to prolonged exposure to saline and humid environments. The proposed process-stabilization technology minimizes quality deviations during repeated production and enables use in reinforcement structures, frames, and other components requiring both structural integrity and environmental durability.

    In electrical and industrial components, such as EV charging structures and industrial equipment frames, pultruded composites offer both mechanical strength and durability. Automated process control facilitates consistent production quality.

    Furthermore, the technology—developed through international collaboration—can be applied to overseas composite manufacturing environments. The recognized and standardized data-driven process management capabilities will provide competitive

	advantages in global markets where reliability and process certification are essential, supporting broader technology dissemination and commercialization. ■ Global Market and International Collaboration Potential Pultrusion process technologies integrated with AI-based recognition and automation have high applicability in international composite manufacturing environments. Standardized process monitoring and data-driven control frameworks established through international collaborative research serve as key competitiveness factors for global market entry.
Rationale for Joint R&D Project	**Critical Technical Barriers** Pultrusion is a representative manufacturing process capable of continuously producing structural composite members across countries and industrial sectors. However, it universally suffers from inherent limitations: multiple process variables—such as roving alignment, fiber tension fluctuations, and resin impregnation uniformity—interact in complex ways, resulting in quality variability across produced parts. In particular, the absence of real-time, quantitative methods to assess quality during processing leads to a reliance on post-process inspection and rework when defects occur. This creates significant constraints in securing structural reliability and in deploying pultruded composites as high-value structural materials. Therefore, it is essential to develop process technologies that can objectively recognize quality states during production and ensure quality consistency in repeated manufacturing. **Why Global Collaboration?** To fundamentally mitigate quality variability in pultrusion, it is necessary to introduce AI-based process recognition and prediction technologies capable of real-time monitoring of process data—such as roving alignment and resin impregnation—and quantitatively interpreting their impact on structural performance. However, such technologies require data analysis capabilities that extend beyond simple image recognition, including correlation analysis among process variables and quality-state determination. The domestic lead institution's existing pultrusion expertise alone is insufficient to independently develop process-data-driven AI algorithms and high-level predictive models, due to technical and resource constraints. This international joint R&D project is optimally positioned for a complementary division of roles: the domestic lead institution will oversee pultrusion process design, process optimization, prototype fabrication, and structural performance evaluation, while the overseas partner institution will lead the development of AI-based process recognition, prediction algorithms, and data interpretation technologies. In this structure, AI is used not to replace process expertise but as an objective tool to support quantitative evaluation and control of process states. International collaboration further enables the accumulation and validation of data under diverse process environments, providing a foundation for objectifying and standardizing quality assessment criteria for pultrusion. This would play a critical role in ensuring the quality reliability and scalability required in the global composites market. The pultrusion process recognition and automation technologies developed through international

	joint R&D can be expanded beyond domestic application and adapted to overseas composite manufacturing environments. By strengthening competitiveness in meeting global requirements for quality consistency and structural performance verification, the project is expected to support broader application and commercialization in fields such as construction, marine structures, electrical systems, and industrial structural components. Therefore, this project is essential for achieving both technical maturity and global competitiveness through international collaboration.
Key Words	1. Pultrusion Process Technology 2. Composite Prototype Quality Uniformity 3. Structural Performance Verification 4. AI-Based Process Recognition 5. Process Automation and Stabilization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	**R&D Objectives** The ultimate goal of this development project is to integrate the AI-based process recognition and data interpretation technologies of the overseas partner institution with the pultrusion process design, prototype fabrication, and performance verification technologies of the domestic lead institution. Through this cooperation, the project aims to establish an “AI-enabled process recognition and automation pultrusion manufacturing platform” capable of improving the quality uniformity of pultruded composite prototypes and quantitatively validating their structural performance. This platform will minimize quality variation during repeated production and ensure the reliability of pultruded composites as structural materials. **Detailed Objectives** 1. Development of Pultrusion-Based Roving Alignment and Resin Impregnation Recognition and Quality Assessment Technologies ● Develop sensor- and vision-based data acquisition systems and AI-based process recognition algorithms capable of real-time monitoring of roving alignment and resin impregnation conditions during the pultrusion process. ● Establish a process quality assessment logic that quantitatively interprets quality states under varying process conditions and distinguishes between normal and abnormal states. ● Secure process recognition technologies capable of detecting early-stage indicators of quality degradation such as roving misalignment and localized under-impregnation to minimize variations in fiber volume fraction and porosity among prototypes.. 2. Development of AI-Integrated Automated Control and Pultrusion Process Stabilization Technologies			

- Develop automated control technologies that link AI-based process quality assessment results to adjustments of key process parameters, including roving tension, resin feed conditions, and pulling speed.
- Improve the traditional operator-dependent process-setting practices to ensure reproducibility and stability of process conditions during repeated production.
- Implement pultrusion process stabilization technologies capable of controlling prototype-to-prototype quality variation within ±5% through AI-integrated process automation.

3. Establishment of Pultruded Composite Prototype Fabrication and Structural Performance Verification Technologies
   - Fabricate structural composite prototypes using optimized pultrusion process conditions and quantitatively evaluate the uniformity of prototype quality.
   - Conduct structural performance tests—including tensile and flexural tests—to compare and analyze quality and performance improvements before and after applying AI-based process recognition and automation technologies.

R&D Target Specifications

Key Performance Indicators	Target Value	Final goal	Evaluation method
Tensile Strength	≥ 10% improvement over baseline	10%	Tensile test (ASTM D638 or ISO 527)
Flexural Strength	≥ 10% improvement over baseline	10%	Flexural test (ASTM D790 or ISO 178)
Fiber Content	Within ±5%	20%	Fiber content measurement via ASTM D2584 and mass analysis
Inter-prototype Property Variation	Within ±5%	20%	Tensile testing of repeatedly manufactured specimens under identical conditions (ASTM D638 or ISO 527)
Roving Alignment Recognition Accuracy	≥ 90%	20%	Accuracy evaluation of AI recognition results based on process imaging data
Impregnation Defect Detection Rate	≥ 90%	20%	Detection success rate compared to actual process data

R&D Plan

1. Research Approach & Strategy
   - This project adopts a process-centered research approach, in which pultrusion process technology serves as the core and AI-based process recognition and automation technologies are applied as supporting tools. The research focuses on analyzing the root causes of quality variation in conventional pultrusion processes, recognizing and assessing quality states in real time during manufacturing, and linking these assessments to process-parameter control to improve prototype quality and structural performance.
   - In particular, the project does not aim to advance AI technology itself; instead, AI is applied

as a practical means to quantitatively interpret and control quality changes occurring during the process. This approach enables both field applicability and reproducibility in repeated production, which constitute the central direction of the research strategy.

2. Key Tasks & Milestones

Year	Key Objective	Major Tasks	Expected Outcomes
Year 1	Process Analysis and Recognition Technology Development	- Pultrusion process analysis - Baseline prototype fabrication - Process data collection and development of foundational AI recognition algorithms	- Secured baseline prototypes - Initial AI-based process recognition established
Year 2	Automation Integration and Process Stabilization	- Advancement of process recognition algorithms - Integration of automated control for key process parameters	- Automated process control integrated - Process stabilization conditions achieved (≥ 90%)
Year 3	Prototype Fabrication and Performance Verification	- Prototype fabrication under optimized conditions - Quality uniformity and structural performance testing	- Performance verification completed - Technology maturity achieved

3. Methodology

- The process recognition technology is implemented by applying AI algorithms to images and sensor data acquired during the pultrusion process. Process-state evaluation is performed by classifying normal and abnormal conditions, and the results are linked to the process-parameter control logic.
- Fiber content is measured through burn-off testing, and structural performance verification is conducted via tensile and flexural tests. The results before and after applying the process recognition and automation technologies are compared and analyzed.
- The performance of the AI-based process recognition system is validated by establishing an internal verification protocol for process data and calculating accuracy and detection rates.

4. Milestones

Research Activity	Year 1				Year 2				Year 3				Notes
Pultrusion Process Analysis and Baseline Prototype Preparation													Existing process analysis
Process Data Collection and AI Recognition Technology Development													Stepwise advancement
Development of AI-Integrated Automated Process													Process-parameter linkage

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	Control													
	Pultrusion Process Stabilization													Repeated-production conditions
	Fabrication of Optimized Process Prototypes													Final prototype
	Quality & Structural Performance Verification													Testing and validation

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**1. Overseas Collaborative Research Institution – AI-Based Process Recognition and Data Interpretation** ● The overseas research institution is responsible for developing AI-based process recognition and data interpretation technologies. This includes designing algorithms that identify and assess roving alignment and resin impregnation conditions during the pultrusion process, and quantitatively classifying process quality states. ● The research scope includes the development of process-recognition algorithms utilizing process imaging and sensor data, as well as establishing logic for distinguishing normal and abnormal process states. The institution will provide the AI development environment and expert personnel, and will supply the domestic lead institution with analysis results such as recognition accuracy and defect-detection rates.
Expected R&R of Korean SME Partner	1. The Korean SME partner – Pultrusion Process, Prototype Fabrication, and Performance Verification ● Technical Contribution and Research Scope ■ The Korean SME partner oversees the overall R&D activities for this project based on its pultrusion process expertise. Its responsibilities include process-condition analysis, process stabilization, prototype fabrication, and structural performance verification. It also ensures the integration and on-site implementation of the AI-based process recognition technology within the actual manufacturing process. ■ The research scope includes pultrusion process analysis, derivation of baseline and optimized process conditions, fabrication of composite prototypes, and fiber-content and structural performance testing. The institution provides pultrusion manufacturing equipment, prototype fabrication infrastructure, and mechanical testing facilities, and is responsible for process automation integration and final technology validation. 2. Applied Technology Commercialization and Mass-Production Verification ● Technical Contribution and Development Tasks ■ The Korean SME partner contributes by conducting pultrusion process analysis, process stabilization, and composite prototype fabrication. It implements automated process control systems that incorporate AI-based process recognition results from the overseas institution, thereby ensuring quality consistency during repeated production.

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	● Testing and Verification Activities ■ For pultruded composite prototypes, the SME performs fiber-content testing and tensile/flexural testing to quantitatively verify the impact of the developed process technologies. When necessary, certified external testing agencies will be engaged to ensure reliability of test results. ● Collaboration Method and Resource Contributions ■ On-site data feedback: The SME provides real production data to Purdue University to enhance the precision of AI-based process recognition algorithms. ■ The SME contributes pultrusion manufacturing equipment and specialized personnel. Under the collaborative framework, the Korean SME is responsible for process operation, prototype fabrication, and testing, while the overseas partner institution handles AI-based process recognition and data interpretation. 3. Inter-organizational Collaboration Framework ■ The domestic lead institution is responsible for pultrusion process execution and prototype fabrication and verification, while the overseas partner institution leads AI-based process recognition and data interpretation. The two institutions operate under a complementary division of roles and jointly share process data, analytical results, and responsibilities for achieving the R&D objectives.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technical Achievements ● Establishment of an AI-integrated pultrusion process recognition and automated quality-control system ■ Developed an integrated process-quality database linking roving alignment, resin impregnation conditions, and key process-variable data. ● • Enhanced prototype quality uniformity and structural reliability ■ Controlled roving misalignment and under-impregnation at the process stage, stabilizing fiber-content variation within ±5%. ■ Minimized structural performance variation between prototypes and secured consistent tensile and flexural performance under repeated production. ● Advancement of pultrusion process technology and validation of structural composite applicability ■ Verified the feasibility of applying pultrusion-based composites as structural components through prototype fabrication and mechanical testing. ■ Transitioned the pultrusion process from experience-based operation to a data-driven, high-reliability manufacturing technology. 2. Commercialization Potential ● Readily deployable manufacturing technology for structural composite markets ■ Enables immediate application in construction, marine, electrical, and industrial structural component markets.

	■ Compatible with existing pultrusion equipment, allowing commercialization without major new equipment investment. ● Differentiated manufacturing competitiveness through AI-driven process enhancement ■ Provides competitive advantage through process-quality recognition and control, beyond simple automation. ■ Improves production repeatability, strengthening competitiveness in securing contracts for structural composite products. ● Cost reduction and profitability improvement through process stabilization ■ Reduces manufacturing defects and rework, lowering overall production costs. ■ Stable quality supports long-term supply contracts and expansion of project-based orders. 3. Industrial and Economic Impact ● Digital transformation of pultrusion manufacturing and enhancement of industrial competitiveness ■ Upgrades SME manufacturing capabilities by shifting pultrusion from experience-based practice to data-driven process control. ■ Contributes to strengthening domestic industry competitiveness through stable production of high-quality structural composites. ● Lightweighting and maintenance-cost reduction through metal replacement ■ Promotes lightweighting, improved environmental durability, and reduced maintenance costs across construction, marine, and industrial structures. ● Strengthened global market competitiveness and export potential ■ Enables domestic companies to meet global standards for quality reliability through internationally validated process-recognition and verification technologies. ■ Supports global market entry and export growth by enhancing pultrusion technology competitiveness.

6. References

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